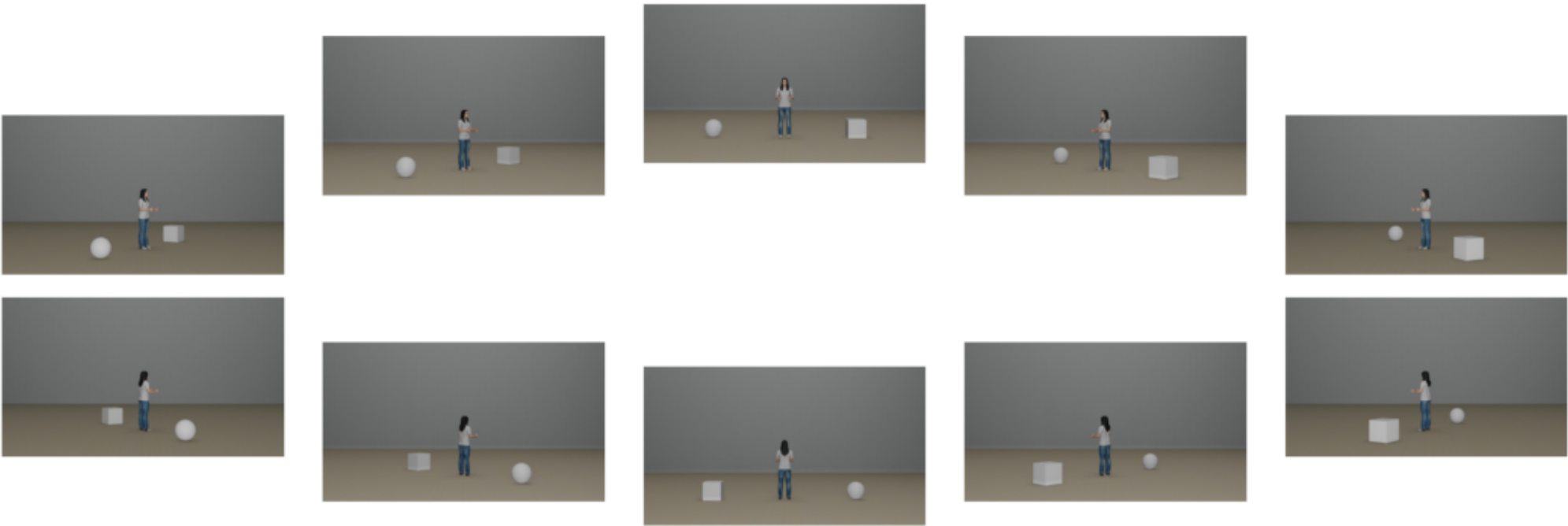


(a)



(b)

Embodiment Tokens



1. Extract keypoints

Diagram showing a person's head (LS) and shoulder (RS) keypoints. The distance between them is $\Delta x = 192 - 172 = 20$ and $\Delta y = 81 - 53 = 28$. The orientation is calculated as:

$$\theta = \arctan2(-(28), 20) \times \frac{180}{\pi} + 360 = 305.55^\circ$$

2. Compute orientation

Token sequence for pose estimation:

<POSE_START> <SHOULDER_L>
<X_172> <Y_53> <SHOULDER_R> ...
<ORIENT_START> <YAW_6> <TORSO_W_2>
<ORIENT_END> <POSE_END>

3. Generate tokens

Rotation Tokens



1. Extract (X, Y) coordinates

OrientAnything

azimuth: 222
 $(180 - 222) \% 360 = 318 \approx 315^\circ$

azimuth: 267
 $(180 - 267) \% 360 = 273 \approx 270^\circ$

2. Extract orientation

Token sequence for rotation:

<OBJECT_START> <OBJ_PERSON>
<X_-4> <Y_6>
<AZIMUTH_315> <OBJECT_END>

<OBJECT_START> <OBJ_FURNITURE>
<X_-98> <Y_38>
<AZIMUTH_270> <OBJECT_END>

3. Generate tokens

(c)

Atomic

Q:
Estimate this person's pose.

A:
<POSE_START> <SHOULDER_L> <X_172> <Y_53>
<SHOULDER_R> ... <ORIENT_START> <YAW_6>
<TORSO_W_2> <ORIENT_END> <POSE_END>

Chain of Thought

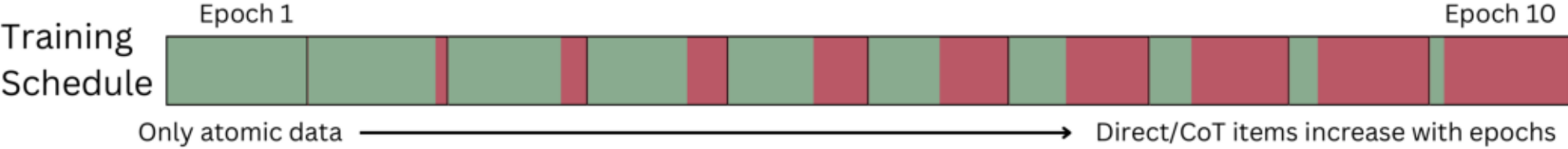
Q:
A person and a chair are shown in the image. Which side of the person is the chair on from the person's perspective? To answer this question, let's think through it step by step...

A:
Left

Direct

Q:
A person and a chair are shown in the image. Which side of the person is the chair on from the person's perspective?

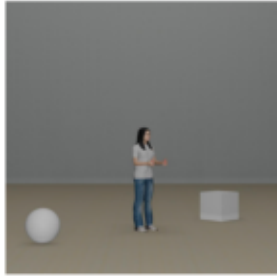
A:
Left



(d)

Synthetic

Naturalistic



Which side of the person is the cube on from the person's perspective?



Which side of the person is the bike on from the person's perspective?



Isle Dataset
Goral et al. 2025



From the perspective of the humanoid minifigure, where is the object located relative to it?



COCO
Lin et al. 2015



Ma et al. 2025



Orientation
Q: Is the stop sign on the left or right side of the man on the bicycle?
A: On the right side.